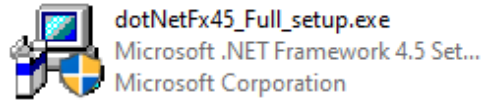
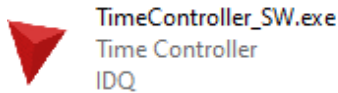


1. Computer software installation (GUI)

In the release package, there are two Microsoft Installer files present; one for the Time Controller software and another for its requirements in the Redistribute folder.



Double-click on the Time Controller SW installer then and follow the instructions.

You may also need to install the dotNetFx45 installer present in the “Redistribuables” folder.

The next steps of this document describe how to update the embedded software (including FPGA firmware) of the Time Controller (mandatory to use the new computer software).

Find out your Time Controller IP address

There are several ways to find out the IP address of the Time Controller. They are listed in this section.

1.1. From the sticker at the back of the device

At the back of the device, depending on the time of manufacturing there is one sticker with either the serial number:



or there with the serial number and the IP address:

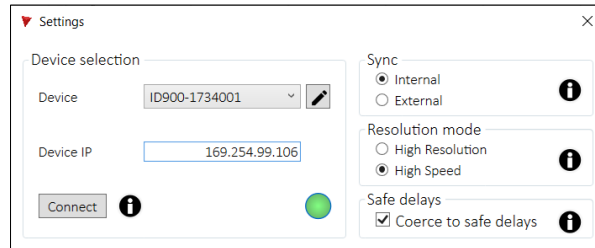


If your device has the IP address, write it down and go to directly to the step 2.

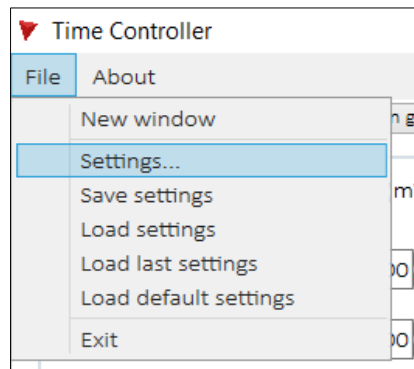
1.2. From the Time Controller GUI Software

Make sure your Time Controller is powered on and connected to your computer through an Ethernet cable (and if needed with the provided USB-to-Ethernet dongle) and then launch the Time Controller software.

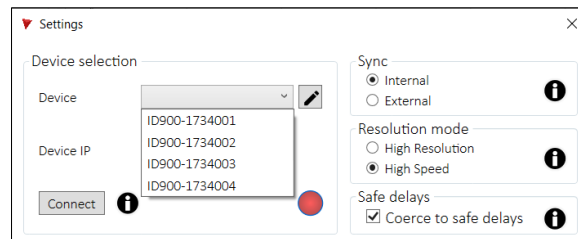
The “Settings” dialog box should pop up.



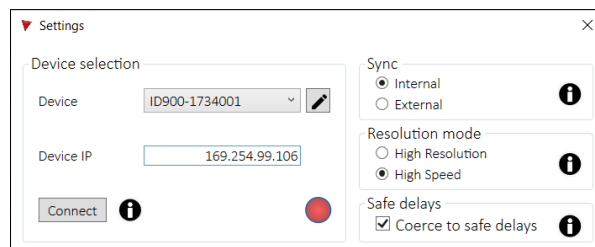
If it is not the case, go in the File -> Settings menu.



In the “Settings” dialog box, click on the drop list next to Device. The serial number of your Time Controller should be listed.



Click on it and the IP address should appear in the field “Device IP”.



Write down the IP address and go to the step 2.

1.3. From the Serial Number

If the IP address does not appear on the sticker at back of the Time Controller, its IP address can be computed from the serial number with the following formula:

Serial Number = AAAA0BBT010

IP = 169.245.XXX.1YY

where

- XXX comes from the AAAA field in the serial number from the following conversion table.

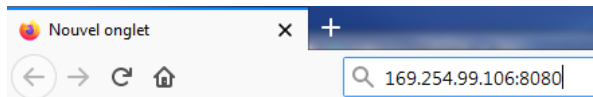
AAAA in serial number	XXX in IP address
1734	99
18xx	100
1908	101

- YY is the device number in the batch (BB field in the serial number)

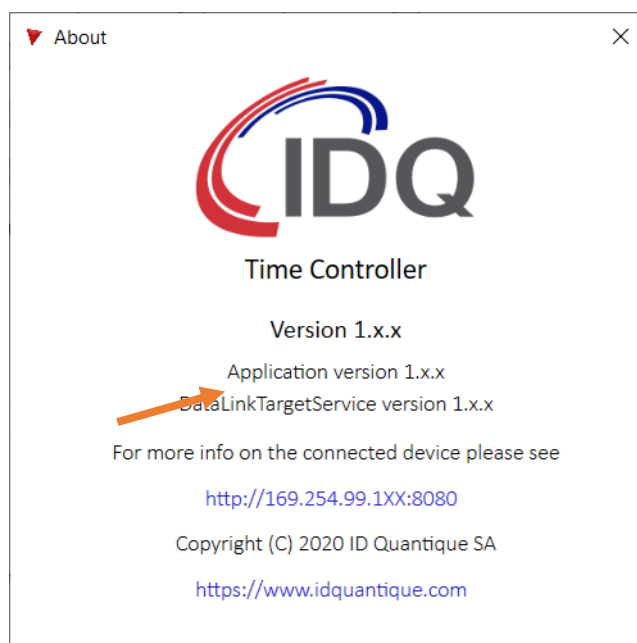
For example, the ID900 with the serial number of 1734006T010 has the IP address “169.254.99.106”.

2. Check the Time Controller version in its web interface

From a web navigator, type the IP address of your Time Controller followed by “:8080”:

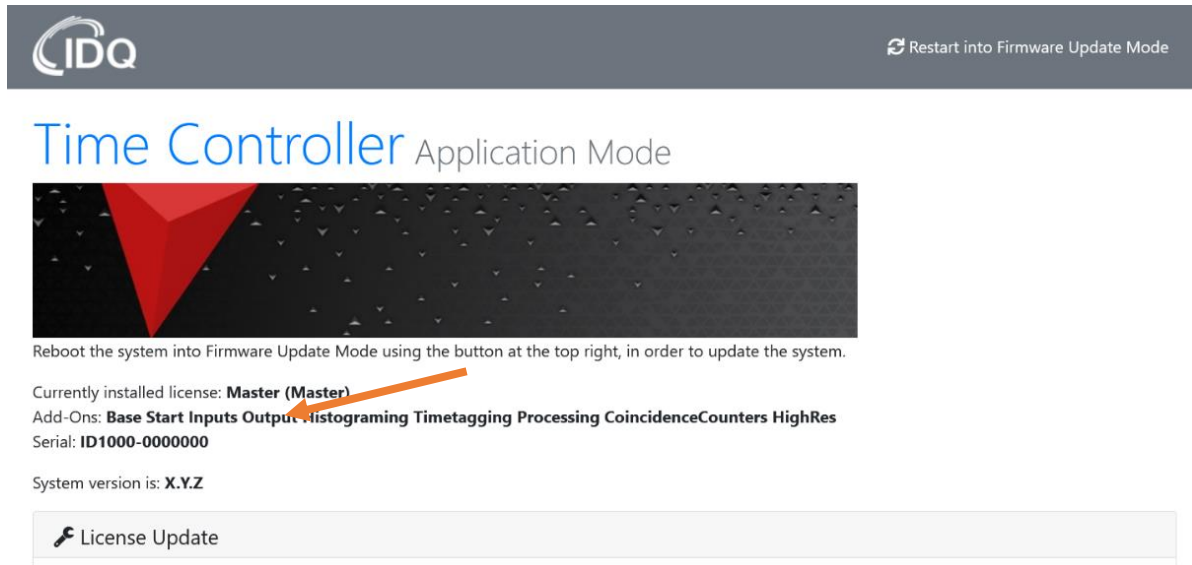


Or, from the Time Controller GUI, open the menu “about” and click on the link displayed in the pop-up window

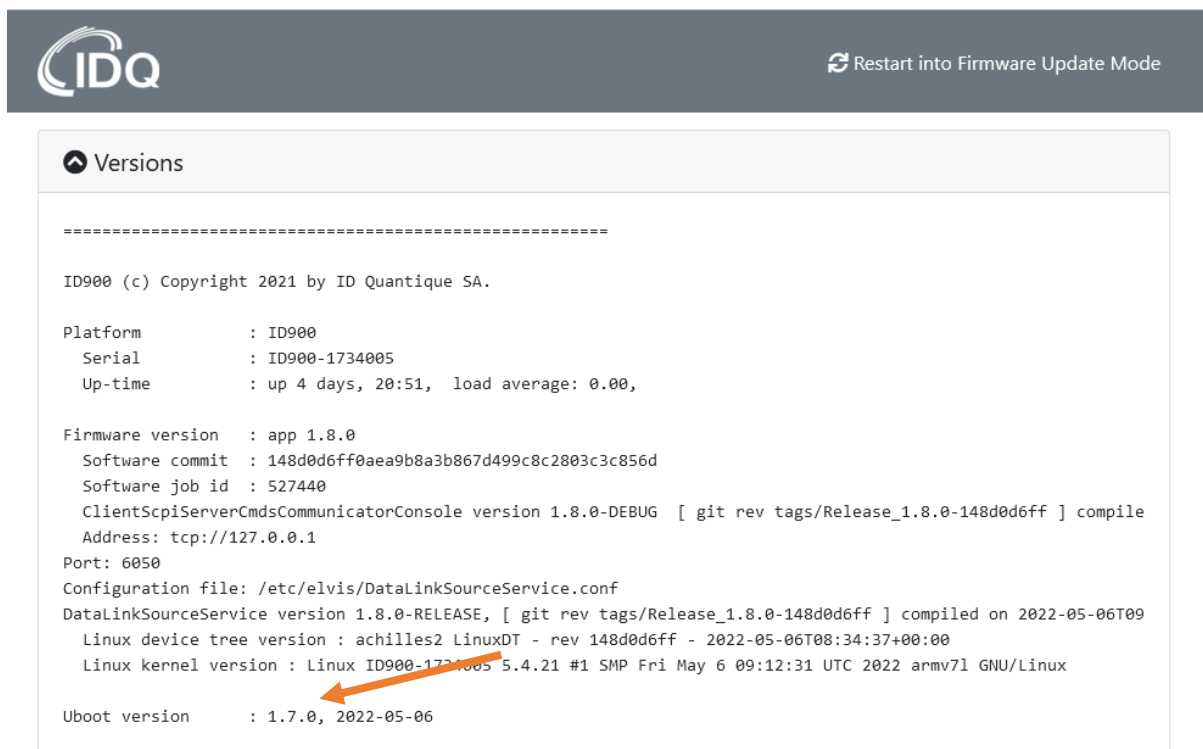


Time Controller update procedure

Check the Time Controller system version (see screenshot bellow).



ID900 only: In the Versions section of the page, check the Uboot version.



3. Updating the Embedded Software

3.1. Files to install

To update the embedded application software, install these files **in this order** (procedure 3.1.1).

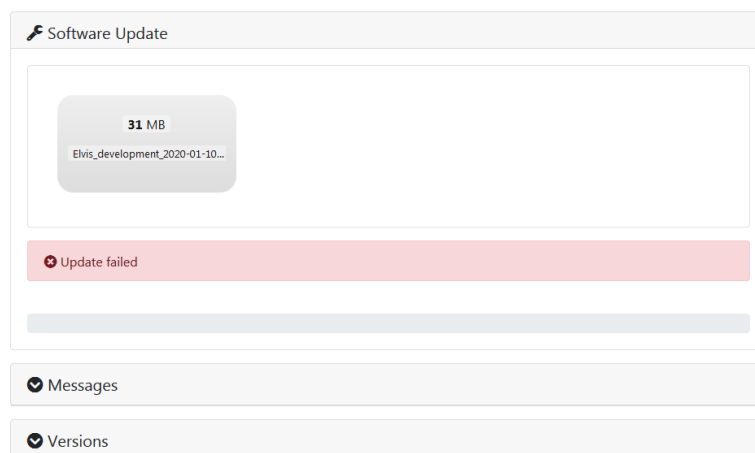
1. ID900 only (install if Uboot version is below 1.7.0): ID900_Recovery_ESW_1_8_0.swu
2. TimeController_Application_ESW_1_10_0.swu

ID900 only: If the following error happen, install ID900_Recovery_ESW_1_4_0.swu before (procedure 3.1.1).



ID-900 Update Mode

Software update for ID-900



Once installed, check the Time Controller System version

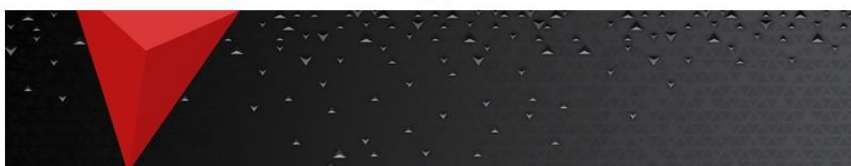
You've now successfully installed the new version of the embedded software on your Time Controller.

3.1.1. How to install a Software Update file (SWU)

Click on "Restart System into Recovery Mode" on the top right of the web page.



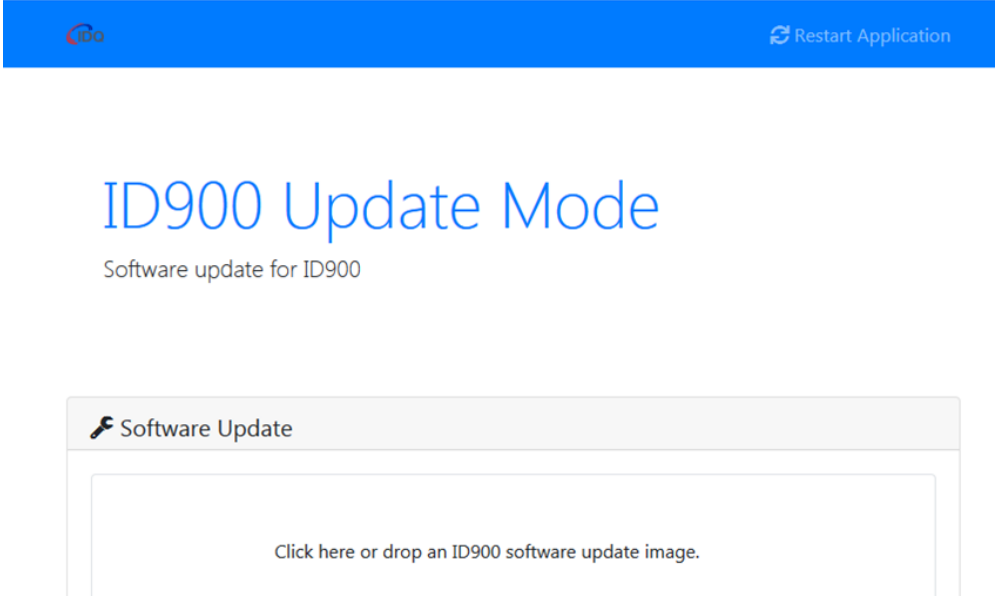
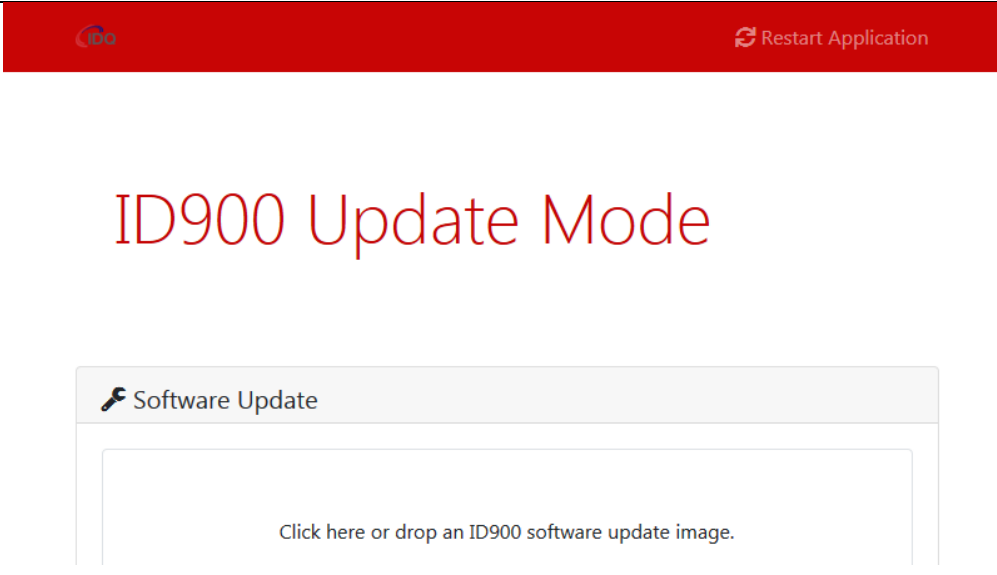
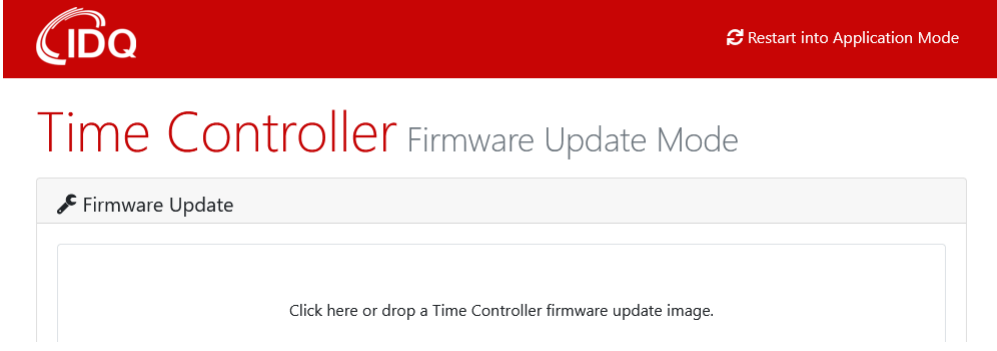
Time Controller Application Mode



Reboot the system into Firmware Update Mode using the button at the top right, in order to update the system.

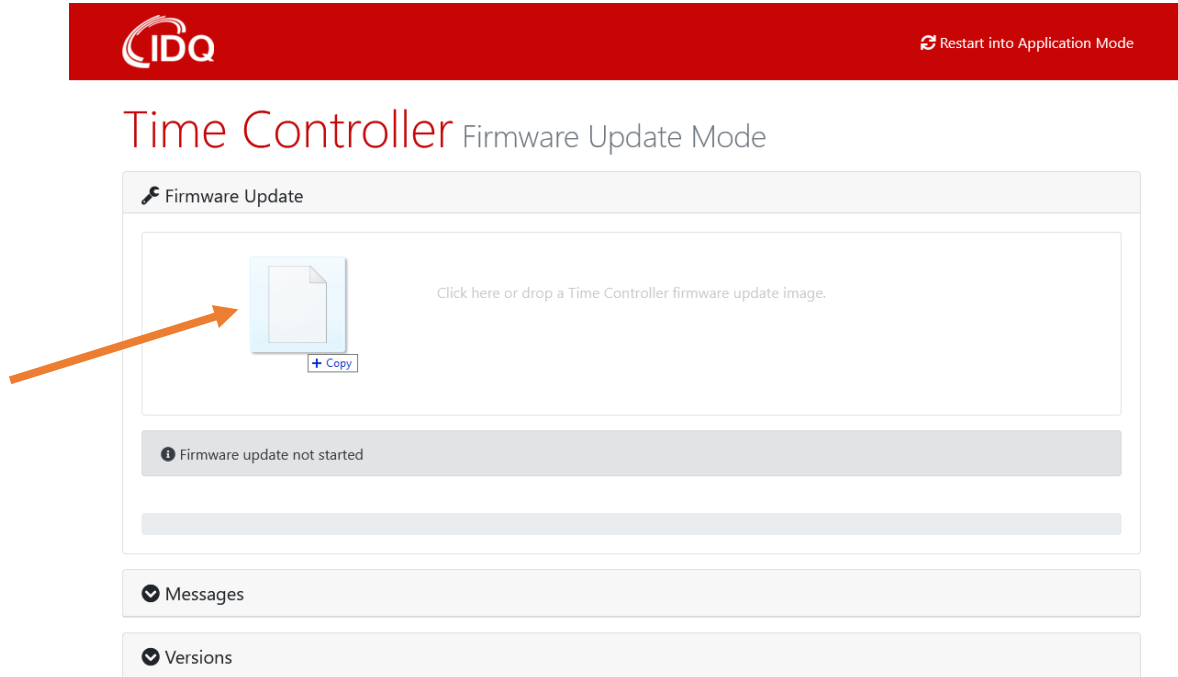
Time Controller update procedure

The message “The system will restart. Please be patient...” will appear. Wait about one minute and the page should refresh by itself (if it does not, refresh manually) and display one of these pages.

Versions	Appearance
< 1.4	
1.4 - 1.7	
1.8	

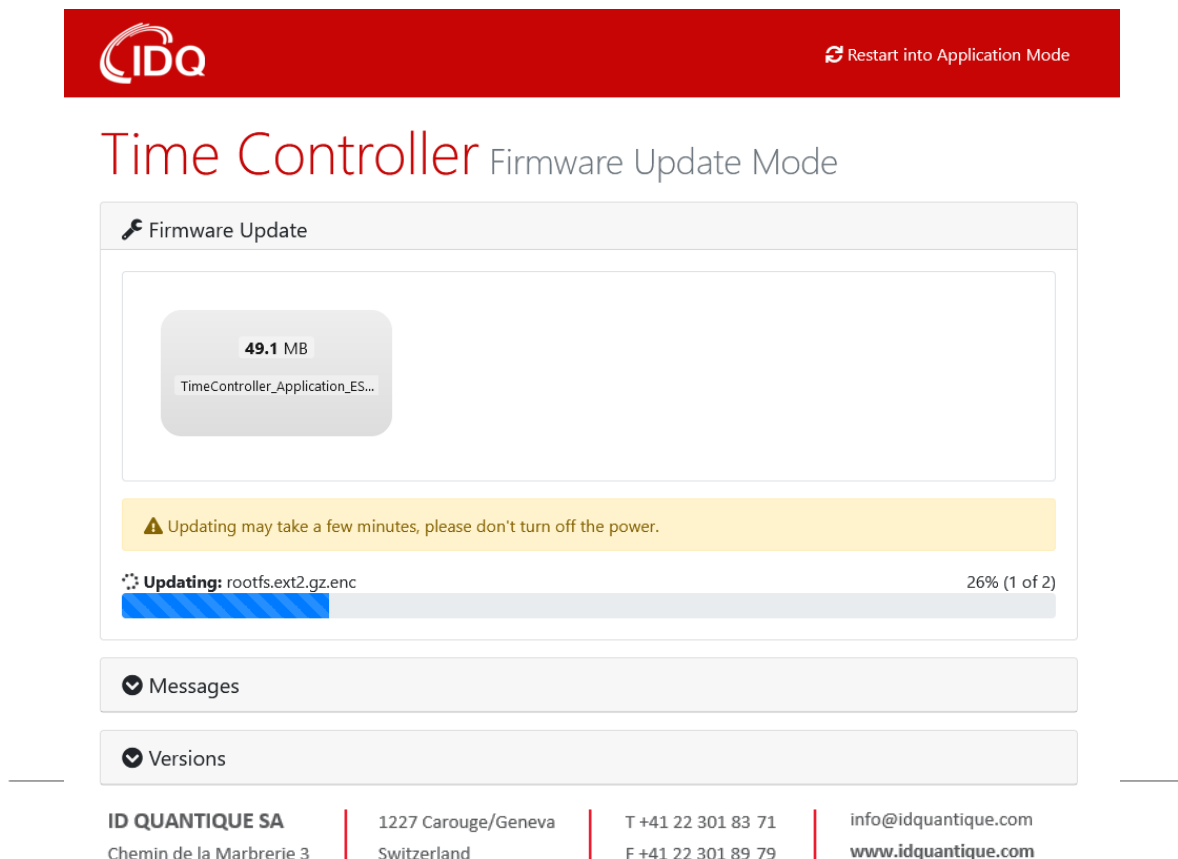
Time Controller update procedure

Click in the "Firmware Update" field or drag and drop the SWU file in the field.



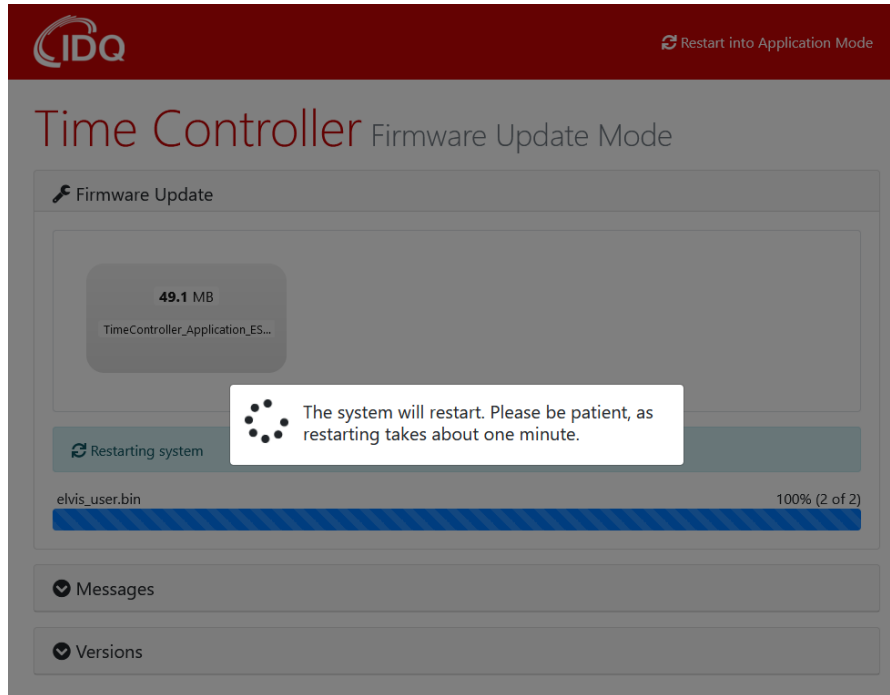
Wait for the progress bar to reach 100% and the system to reboot. (Do not worry, the bar will remain at 0 % for one or two minutes).

DO NOT POWER OFF THE TIME CONTROLLER DURING THIS STEP



At the end of the update, a popup with “The system will restart. Please be patient...” will appear and the Time Controller will reboot with the new embedded software.

Note – the Time Controller will reboot in ‘Application’ mode. To install another SWU file it’s necessary to switch back to ‘Update’ mode, as described earlier.



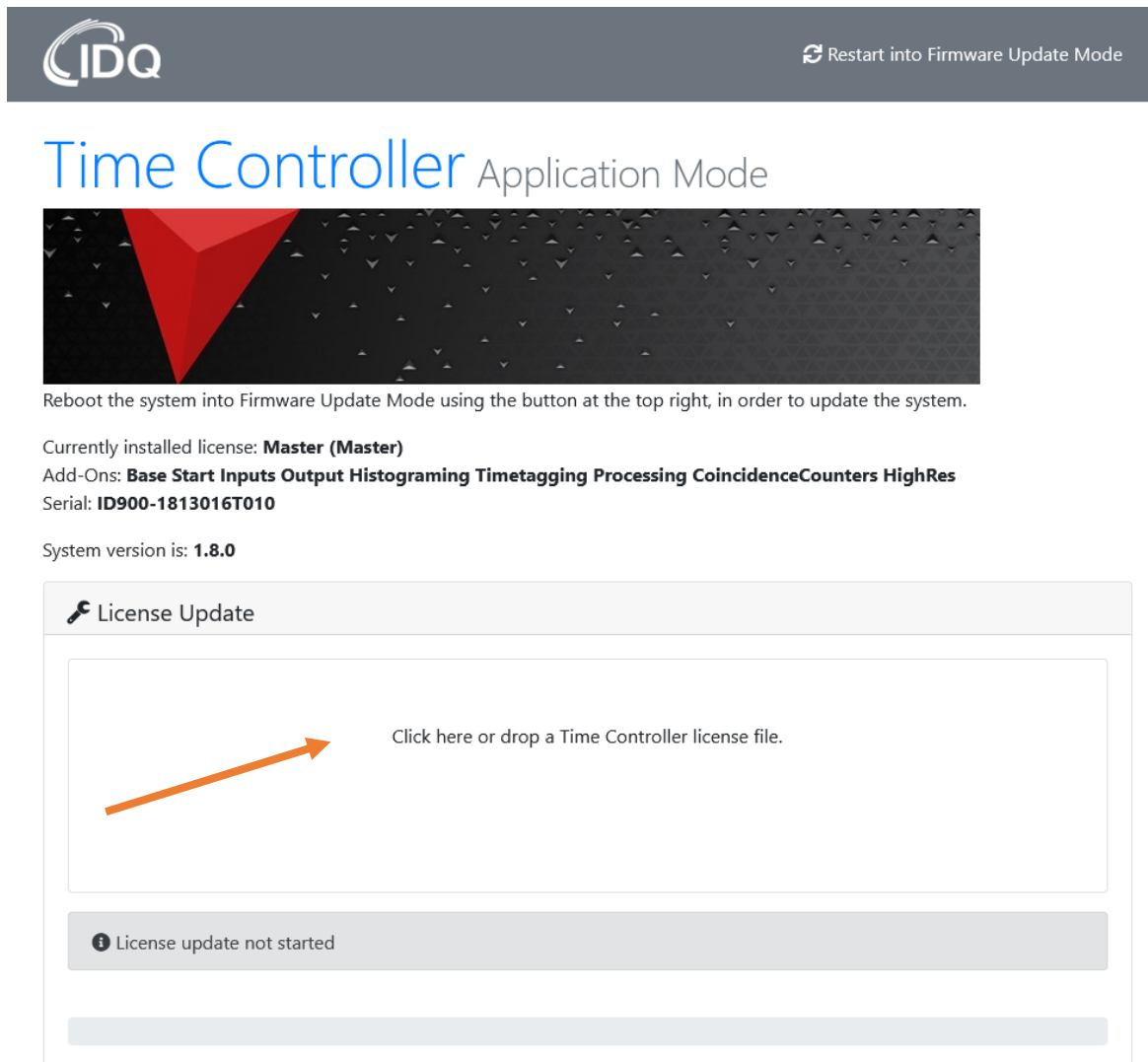
Wait for the popup to disappear. If the message is still shown after 2 minutes or if the device does not actually restart, power off the Time Controller by pushing on the power button for more than 5 seconds and then power on again the device. Then wait about one minute and refresh the page.

4. Install a license

No further license installation is required upon reception of the Time Controller, for it is shipped with the purchased license already installed. The license remains active after a firmware update.

It is possible to upgrade the Time Controller license. Then, you will need to install the new license received from ID Quantique.

Open the Time Controller web interface (see step 2) and drag the license file in the “License Update” area.



Refresh the Time Controller web interface to update the “Currently installed license” label.

You are now ready to use your Time Controller.